IS-RoadDet: Road Vector Graph Detection With Intersections and Road Segments From High-Resolution Remote Sensing Imagery

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***Abstract*— Extracting road vector graphs with high accuracy from high-resolution remote sensing imagery presents a significant challenge. The prior end-to-end algorithms have typically modeled the road graph as a general graph structure with vertices and edges, denoted as $G = (V, E)$, which is a standard approach in graph generation tasks. However, the traditional $G = (V, E)$ graph representation with vertices and edges, utilizing very small edge units, can conflict with the road network's geometric structure and the inherent features of road instances, leading to issues such as false positives and disconnected roads. In this article, the IS-RoadDet framework is proposed to generate a road vector graph with intersections (I) and road segments (S), denoted as $G = (I, S)$, which leverages the minimum road topology unit features of road to improve road topology. Compared to $G = (V, E)$, instead of detecting a large number of vertices to maintain the road topology connectivity, $G = (I, S)$ uses road segments with minimum road units to avoid false positives. In IS-RoadDet, the intersection and road segment detector (ISDetector) is introduced to detect intersections and road segments as independent instance objects with joint learning, and an intersection connectivity strategy (ICS) is designed to establish connectivity between road segments with intersections. Empirical experiments conducted on the SpaceNet3 and Sat2Graph datasets substantiated the superior performance of the proposed road segment modeling method. The code will be made available at: https://github.com/WanderRainy/IS-Road and http://rsidea.whu.edu.cn/is-road.htm.**

***Index Terms*— Instance detection, intersection, remote sensing images, road segment, road vector graph detection.**

I. Introduction

THE primary task of extracting roads from high-resolution remote sensing imagery is to capture the spatial information of the road network. Road vectors not only encompass the spatial information of roads but also provide a precise representation of the topological structure of roads, including complex scenarios such as overpasses. Furthermore, owing to the ease of conducting statistical analysis on road vectors, they can be directly applied in navigation, urban planning, and emergency response, among other domains [1], [2]. Road vector data are heavily dependent on the information gathered from professional surveying vehicles and manual inspection and correction processes. However, this approach is both costly and time-consuming, presenting challenges in keeping up with the timely updates required for large-scale road networks.

The existing methods for road vector extraction can be categorized into post-segmentation methods and vertex-based methods. In recent years, advancements in the fundamental semantic segmentation tasks [3], [4], [5], [6], [7] have contributed to the enhancement of road segmentation results. A common method for deriving a road vector graph from segmentation results involves the application of postprocessing techniques, such as morphological operations and heuristic algorithms [8], on skeletonized segmented roads. However, the challenges inherent in the road segmentation task, such as occlusions and intersections, can disrupt the connectivity of the resulting road graph, as illustrated in Fig. 1(a), where $G = P^S$ represents the road vector graph derived through postprocessing (P) on the segmentation results (S) [3], [9], [10], [11]. In addition, road vector graphs generated from pixel-wise road extraction lack the necessary topological structure to accurately represent road overpasses [12], [13]. To address this issue, recent research has shifted toward direct road vector graph generation using the vertex-and-edge model, denoted as $G = (V, E)$ [12], [13], [14]. The iterative methods [2], [13], [14], [15] predict the next adjacent point from the current point to establish connected road edges, while the one-shot methods [16], [17], [18], [19] extract all the vertices simultaneously and then determine the connectivity of adjacent vertices. Mastering [2] adopts a reinforcement learning approach to deal with misalignment between model predictions and intended purpose and sequentially adds new edges. SAM-Road [20] first predicts the graph vertices and determines the connectivity to obtain edges by transformer decoder. However, the vertex-and-edge model for connectivity primarily relies on adjacent vertices and local information, neglecting the connectivity of topological vertices and long-distance road segment object features. Each road vertex is not only related to its neighboring road vertices but also to all the vertices on the same

Received 17 July 2024; revised 23 August 2024; accepted 17 September 2024. Date of publication 17 October 2024; date of current version 2 December 2024. This work was supported by the National Natural Science Foundation of China under Grant 42325105 and Grant 42071350. *(Corresponding author: Yanfei Zhong.)*

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Digital Object Identifier 10.1109/TGRS.2024.3483113

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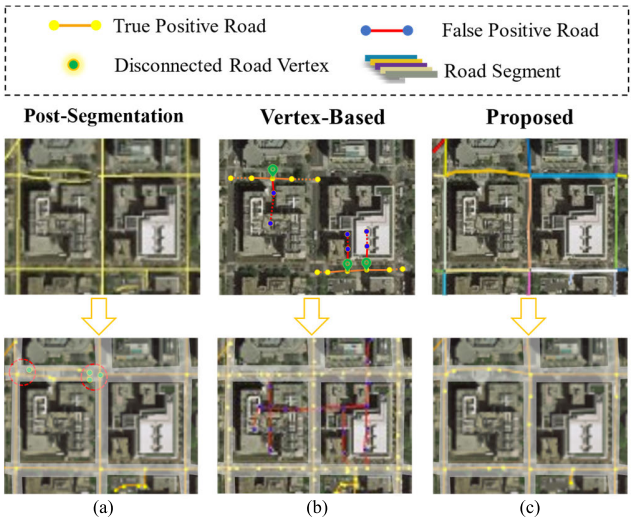


Fig. 1. Visual illustrations of different road vector graph models. In contrast with prior work, the road graph structure is split into road segment objects instead of post (P)-segmentation (S) and vertices (V)–edges (E). The green dots refer to disconnected road vertices caused by the occlusion problem, while the blue dots and red edges refer to false positive roads in the complex background. The light white is ground truth road. (a) $G = P^S$. (b) $G = (V, E)$. (c) $G = (I, S)$.

road segment. In complex backgrounds, this greedy modeling approach, relying on only a small portion of the road, can lead to a significant number of false positive roads, as depicted in Fig. 1(b).

In this article, to solve these problems, the IS-RoadDet framework is proposed to generate a road vector graph by detecting the road segments (S) and intersections (I), $G = (I, S)$. The IS-RoadDet framework is made up of the intersection and road segment detector (ISDetector) and the intersection connectivity strategy (ICS). In this study, to gain a more detailed understanding, the structure and elements of the road vector graph were redefined.

In the proposed approach, the vertices on the road vector graph are classified into four groups: intersections, endpoints, corner points, and trivial points, as illustrated in Fig. 2. Intersections and endpoints play a crucial role in governing the topology of the road vector graph and possess clear spatial location attributes, while corner points and trivial points do not impact the topology. From this perspective, the previous methods have confused topological vertices and nontopological vertices during modeling. Furthermore, edges that are too small between nontopological vertices can result in false positive roads [13], and the greedy segmentation of nontopological vertices can lead to disconnected roads when not distinguishing the topological vertices. To address this issue and distinguish between topological and nontopological vertices during modeling, the road segment (S) is redefined as having two topological vertices as endpoints, with multiple nontopological vertices connecting them internally, which is the minimum road topology unit. This approach allows us to decompose the road vector graph into road segments, as illustrated in Fig. 2, where the connectivity between road segments is controlled by the topological vertices, namely,

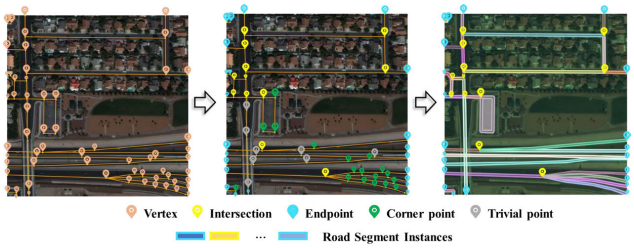


Fig. 2. Representation of the model used to split the road network into intersections and road segments. The four types of colored points represent the four types of road vertices. Different colored roads represent different road segment instances.

intersections (I), while the connectivity within road segments is determined by the nontopological vertices. Based on these concepts, the road vector graph is modeled as $G = (I, S)$. Within this IS-RoadDet framework, the road segments and intersections are treated as independent object instances. In the ISDetector, a query-based instance extraction method along with a semantic segmentation architecture is employed to obtain the two elements, i.e., intersections and road segments. The ICS combines the connectivity of topological and nontopological vertices to generate the road vector graph in a one-shot manner.

In this article, we propose the IS-RoadDet framework, a method for road vector graph detection, which aims to address false positive and disconnected roads by the minimum road topology unit model, $G = (I, S)$. Our main contributions can be summarized as follows.

1) The IS-RoadDet framework is proposed to generate a road vector graph with $G = (I, S)$ from VHR remote sensing imagery. The IS-RoadDet framework first represents the road topology based on the minimum road topology unit, effectively distinguishing road topological points from nontopological points in the different modeling units to balance the precision and connectivity of road vector graph generation.
2) The ISDetector is proposed to obtain the road segments (S) and intersections (I) of roads. It considers the spatial correlation of road segments and topological points and uses joint learning to optimize road segment detection and intersection segmentation.
3) The ICS is introduced to combine road segments and intersections into a road vector graph. Meanwhile, the ICS considers the minimum connectivity property of the nontopological vertices in the road segments to mitigate the occlusion problem.

The rest of this article is structured as follows. In Section II, the related works on road vector graph generation are presented. In Section III, a detailed overview of the new road graph modeling approach, and the specific implementation details of the IS-RoadDet framework are presented. In Section IV, the datasets, the implementation of the experiments, and the presentation and discussion of the results of the experiments are provided. Section V provides our conclusion and the limitations of the present work.

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II. Related Work

A. Post-Segmentation Methods

Extracting pixel-wise road information from high-resolution remote sensing imagery constitutes a longstanding task within the domain of the typical binary semantic segmentation tasks. The pixel-wise road segmentation results contain crucial information regarding road locations, and a prevalent practice involves directly deriving the road vector graph through the utilization of skeletonization and subsequent postprocessing.

In the context of road segmentation tasks, recent research has predominantly tackled the challenges associated with occlusions induced by dense trees and high-rise buildings. These challenges often lead to disconnections in road extraction results, particularly concerning the slender shape attribute of roads, as illustrated in Fig. 1(a). The disconnected roads from pixel-level segmentation can lead to a disconnected road vector graph after postprocessing. To improve the accuracy of road segmentation results, many researchers have incorporated explicit or implicit modeling of the prominent shape features of roads into supervised learning. DinkNet [21] pioneered the use of dilated convolution to implicitly model the long-distance features of roads, securing victory in the DeepGlobe Road Extraction Challenge [22]. GAMSNet [10] models the long-distance features based on multiscale and attention mechanisms, while CoANet [11] and DSCNet [23] explicitly learn the slender and tortuous shape features of roads, thus enhancing the road segmentation tasks. The method developed by Batra et al. [9] uses joint learning of orientation and segmentation to improve road connectivity, and the topology-aware loss term [24] supervises the road segmentation results based on road connectivity.

Postprocessing optimization of road segmentation results is aimed at achieving improved connectivity [8], [25], [26]. Some algorithms are designed based on the statistical laws of road structure to heuristically enhance the results. However, the modeling goal of the road segmentation methods is to identify all the trivial points on the road surface, rendering the occlusion problem more challenging than in the road vector graph generation task and increasing the disconnected road problem. Moreover, heuristic postprocessing is challenging to apply to roads with complex morphological structures, such as overpasses.

B. Vertex-Based Methods

The road vector graph generation task seeks to directly derive the representation of a road vector graph from remote sensing imagery. Employing the graph representation model $G = (V, E)$, the existing methodologies can be categorized into iterative and one-shot approaches based on the order of extracting vertices and edges.

The iterative approach involves the sequential extraction of vertices and edges, progressively connecting them to form a complete graph, emulating the manual annotation process. RoadTracer [14] pioneered the iterative approach, achieving the direct generation of a road vector graph. Subsequent works have refined the acquisition of vertices and edges [2], [13], [15], [27], [28], [29].

In contrast, the one-shot approach extracts all the required points and edges simultaneously, constructing the road vector graph in a single pass. Sat2Graph [12] was the first method to accomplish one-shot generation of edge embeddings. Subsequent studies focused on enhancing the acquisition and representation of edges [1], [16], [17], [18], [19], [20], [30]. However, these primitive graph modeling methods do not distinguish the significance of different vertices, resulting in the generation of many trivial points to ensure connectivity while causing numerous false positive roads.

C. Instance Detection

Different road segments with different geometric attributes are the same object class and different instances. A review of the representative instance extraction methods reveals three primary classes: region-based methods, instance activation-based methods, and query-based methods. In the region-based category, Mask R-CNN [31] serves as a benchmark, detecting object bounding boxes (bboxes) via a region proposal network (RPN) and subsequently extracting the classification and mask information based on regions of interest (RoIs). This two-stage instance extraction methodology has set the standard for research tasks in this domain [32], [33], [34], [35]. Representing the instance activation-based techniques, SOLO [36], [37] identifies the targets and achieves segmentation through pixels that carry meaningful information [38], [39]. As one of the first query-based methods, DETR [39] has demonstrated that a convolutional backbone with an end-to-end set prediction-based transformer encoder–decoder can yield a robust performance [40], [41], [42].

III. IS-RoadDet

To solve the problem of false positive roads generated by meaningless vertices based on $G = (V, E)$, the IS-RoadDet framework is proposed to generate a road vector graph with intersection (I) and road segment (S) as $G = (I, S)$. Fig. 3 illustrates the overall framework, which comprises two key steps.

1) *ISDetector:* This step involves acquiring two crucial elements: the road segments and intersections.
2) *Intersection Connectivity Strategy (ICS):* In this step, the connectivity of the road segments and intersections is integrated to generate the road vector graph.

In contrast to the conventional modeling approach of $G = (V, E)$, the proposed model $G = (I, S)$ involves two elements: intersections and road segments. Unlike the previous methods that only require vertices and their connectivity, the proposed modeling approach necessitates the extraction of road segments and intersections. While intersections can be readily obtained as distinct targets through segmentation, extracting road segments requires instance-level processing due to the varying geometric attributes. Road segments can be represented by ordered point coordinates or by single-pixel-wide centerlines. Given that the diverse curvatures often exhibited by road segments, pixel-wise centerlines have been shown to be superior in capturing geometric structures compared to discrete points. Consequently, in the proposed approach,

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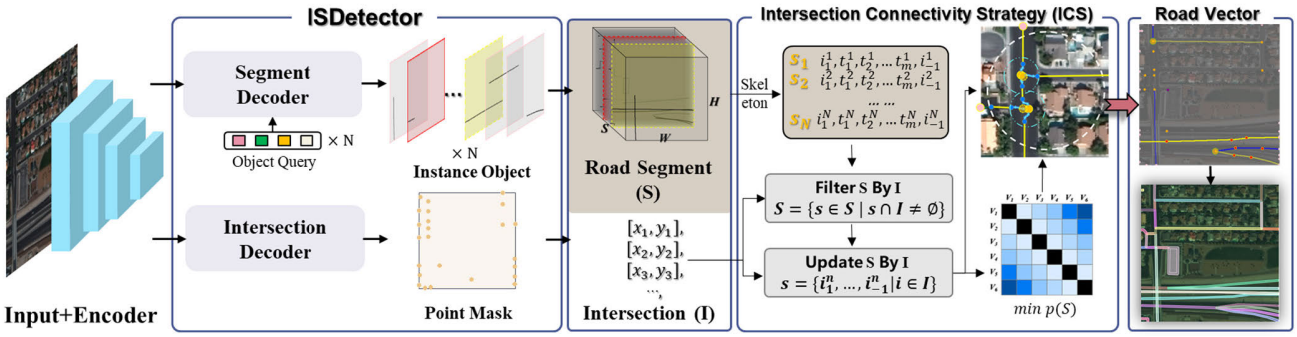


Fig. 3. IS-RoadDet framework with its two main components. Given an input image, the ISDetector extracts the intersections (I) and road segments (S). The ICS combines the two elements to generate the road vector graph and improve the connectivity within the road segments.

an instance segmentation algorithm is employed to extract the road segments, based on the connectivity of all the nontopological vertices. Intersections, on the other hand, encompass the connectivity of all the topological vertices. Recognizing a robust positive correlation between the road segment instance segmentation and intersection extraction tasks, the ISDetector is employed for joint learning to enhance the performance for both tasks. Combining these two elements results in the generation of a comprehensive road vector graph. The connection between road segments relies predominantly on the location information of intersections. To optimize the connectivity within road segments and address the occlusion-related segmentation challenges, the ICS is designed to enhance the connectivity optimization within road segments.

A. Road Graph With Intersection (I) and Road Segment (S)

A road vector graph is modeled as $G = (I, S)$. Each endpoint of a segment, as intersection $i_j \in I$ (j is the index of intersection of road vector graph) represented by coordinates $[x_j, y_j]^{\mathrm{T}}$, defines the road network structure, as illustrated in Fig. 2. Intersections and endpoints with $\lambda_d \neq 2$ determine the topology of the road vector graph, represented by the set of topological vertices I. Corner points and trivial points with $\lambda_d = 2$ are considered nontopological vertices. Here, λ_d denotes the number of edges connected to a vertex. Each $s_k = \{i_j, t_1, t_2, \ldots, t_m, i_{j+1}\} \in S$ represents a road segment determining the connectivity of the nontopological vertices t_m. A probability distribution over roads is generated by following a simple process: 1) sampling a set of topological vertices, i.e., intersections, as I and 2) sampling a set of road segments S that connect them. Formally, for an image I_{image}, the road vector graph G is derived as follows:

$$\begin{aligned} G = (I, S) &= \arg\max_{i,s} \sum P(i^{\text{pair}}, s | I_{\text{image}}) \\ &= P(S|I, I_{\text{image}}) P(I|I_{\text{image}}). \end{aligned} \tag{1}$$

As road segments are the smallest topology units in the road vector graph, the joint distribution is factorized as a product of the conditional distributions based on the assumption that the road segments have independent and uncorrelated distributions

$$P(s_1 \cap s_2 \cap, \ldots, \cap s_m | I, I_{\text{image}}) = \prod_{j=1}^{m} P(s_j | I, I_{\text{image}}) \tag{2}$$

where s_j denotes the jth road segment and m is the total number of road segments.

B. ISDetector

The ISDetector is a joint learning model that concurrently extracts instances of road segments and road intersections. The two ends of each road segment are the location of intersection points, and the two strongly correlated tasks can achieve better results through the use of joint learning. This simultaneous extraction can be advantageous in leveraging the symbiotic relationships between the two tasks, leading to enhanced semantic feature representation.

1) Road Segment Detection: Road segments are represented as instance objects. The distinctiveness of road segment instances compared to other instances lies in their slender shape characteristics, mutual adherence, and notable scale diversity with a large aspect ratio. Employing a region-based framework for road segment detection, a pyramid of feature maps is utilized, generated by a feature pyramid network (FPN), with varying sizes and a fixed number of channels (usually 256-D). This adaptation accommodates the scale diversity inherent in road segments. Given the long-distance slender nature of roads and geometric distribution of different road segments, the instance activation-based approach is not applicable for road segment detection.

The region-based two-stage road extraction approach can perform well in regularly distributed roads, but two problems arise with complex road shapes and road layouts, as shown in Fig. 4. The localization detection box of a long-curved road often covers multiple road segments, and region-based instance segmentation is based on the image patch in the detection box. Therefore, multiple road segments will be segmented into the same road instance. Similarly, the same road will be covered by multiple detection frames, which will also result in the same road segment being segmented multiple times in the different detection frames. This phenomenon leads to the region-based approach being unable to handle this complex situation. To overcome the problem caused by detection boxes, the query-based instance detection approach is used to circumvent this problem. The query-based approach can directly query the corresponding mask, while the extraction of multiple road segments considering with each other, thus avoiding a road segment being detected multiple

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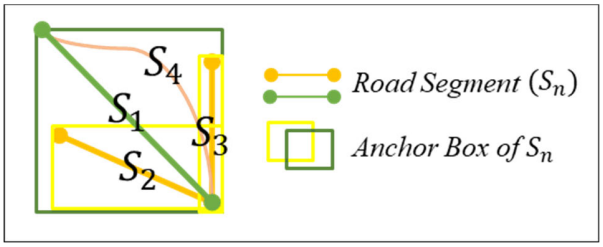


Fig. 4. Complex geometry and distributions are not suitable for box positioning with the ISDetector. The location detection box for a long road segment often covers multiple short road segments, as the anchor box of S_1 contains anchor boxes of S_2 and S_3. Complex road structures result in two road segments that may have the same anchor box, as in S_1 and S_4.

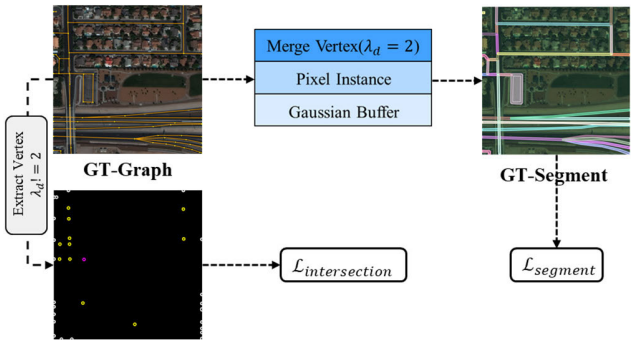


Fig. 5. Training labels for ISDetector generation. The road segment instance labels and intersection masks from the road vector annotations in any format are generated ISDetector training.

times or two road segments being detected in the same road segment.

2) Intersection Detection: The intersection detection task is aimed at identifying the topological vertices within the road vector graph, encompassing intersections and endpoints, as illustrated in Fig. 3. Most of the previous point extraction methods do not distinguish between topological and nontopological vertices and directly greedily extract all the vertices [2], [12], [17], [19]; however, the proposed approach distinguishes between topological and nontopological vertices. The conventional method of distance interpolation for obtaining supervised signals from trivial points leads to inaccuracies due to the random distribution of nontopological vertices in road segments. This method is also a primary contributor to false positive issues. In comparison, the proposed method of distinguishing topological vertices is simpler and more effective, benefiting from their clear topological and spatial location attribute features. Intersection detection is designed to focus on the topological features of roads. By employing joint learning with road segment detection, road symbiosis information is leveraged to enhance the overall performance. The architecture of intersection detection, as depicted in Fig. 3, shares the feature encoder weights with the road segment detection. It utilizes feature maps from the FPN to cascade into a semantic segmentation head. The intersection detection prediction head produces a 1-D heatmap indicating the location of the topological vertices.

3) Training ISDetector: Road vector graph labels are commonly provided in the form of minimum vertex pairs, such as $[(v_1, v_2), (v_2, v_3), \ldots, (v_{m-1}, v_m)]$ [12]. To facilitate

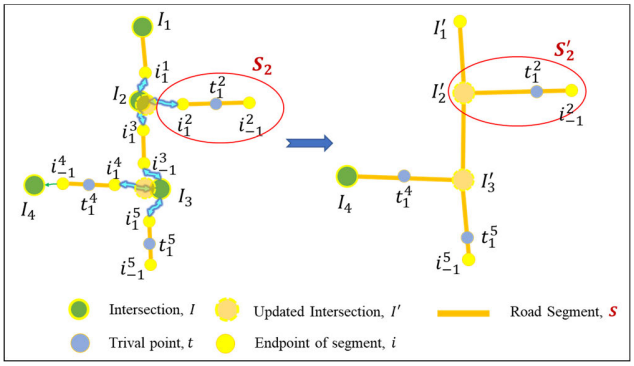


Fig. 6. Illustration of intersection optimization and updates to the road segments in the ICS. Each road segment instance (orange lines) has two endpoints (yellow dots), which are used to optimize the primitive intersection location (green dots), and is updated by the optimized intersection (light yellow dots).

supervised training for road segment detection, the road vector graph was transformed into individual road segment instances through the following steps: 1) merge edges with vertices ($\lambda_d = 2$) into road segments in linestring form, represented as $([x_1, y_1], [x_2, y_2], \ldots, [x_n, y_n])$; 2) rasterize the linestring form of road segments into pixel centerline form, denoted as $[x_i, y_i], [x_{i+1} = x_i \pm 1, y_{i+1} = y_i \pm 1]$; 3) apply a buffer (usually 3 pixels in width) to the pixel centerline form; and 4) convert the pixel centerline form of road segments into CoCo instance object form. ISDetector undergoes end-to-end training by aggregating the losses from both the road segment detector and intersection detector: the road segment detection loss $\mathcal{L}_{\text{segment}}$ is the same as that in [40] and the intersection detection loss $\mathcal{L}_{\text{intersection}}$ is single-class binary cross-entropy loss, as shown in Fig. 5.

C. Intersection Connectivity Strategy (ICS)

After obtaining the intersections and road segments, the ICS is introduced to seamlessly integrate these two elements and generate the road vector graph. The connectivity encompasses topological vertices from intersections and nontopological vertices from road segments. The ICS pipeline, as depicted in Fig. 3, takes the intersection heatmap and road segment instances as input, producing the road vector graph as output. Upon receiving the intersection heatmap, the process is initiated by extracting the local maximum points through nonmaximum suppression (NMS) across all the nonbackground classes, resulting in primitive intersections denoted as $I^P \in \mathbb{R}^n$. Subsequently, for each road segment instance, its validity is assessed based on whether it contains primitive intersections or not, determined by $S \cap I^P$. Invalid road instances are discarded, and valid road instances undergo skeletonization. The Douglas–Peucker (DP) algorithm [39] is used to establish the connectivity of the nontopological vertices in the road segments, denoted as $S = \{v_1^i, v_2^i, \ldots, v_m^i\}$. Notably, the location of primitive intersections is optimized, as shown in Fig. 6, by combining the nontopological vertices at segment ends within the original intersection buffer. Finally, the optimized intersections are utilized to update the segment end vertices as topological vertices, and the combined

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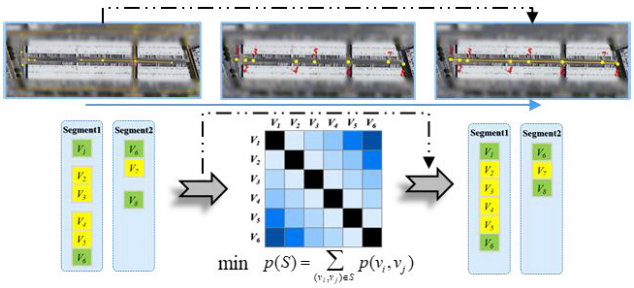


Fig. 7. TSP refine of road segments in the ICS. The shortest distance property of the road segments is used to optimize the connectivity of the nontopological vertices within each road segment.

connectivity of intersections and road segments generates the road vector graph

$$I = (x_I, y_I) = (x_{\mathrm{PI}}, y_{\mathrm{PI}}) + \sum_{i=1}^{N} \frac{N}{D_i} \times (\Delta x_i, \Delta y_i)$$
$$= (x_{\mathrm{PI}}, y_{\mathrm{PI}}) + \sum_{i=1}^{N} \frac{N}{D_i} \times ((x_{\mathrm{PI}}, y_{\mathrm{PI}})\text{-}(x_i, y_i)) \quad (3)$$

where PI is the primitive intersection, I represents the optimized primitive intersection, N is the number of nontopological vertices in the buffer of the intersection, and D is the Euclidean distance between the nontopological vertex and primitive intersection.

The road segment detection algorithm potentially leads to topological errors and disconnections within road segments because the connectivity of nontopological vertices is contingent on the instance segmentation of road segments. An intriguing observation is that upon splitting the road network into road segments, the nontopological vertices within each segment naturally connect through the shortest total distance. Formally, the connectivity of each nontopological vertex v_m^i in road segment $S = \{v_1^i, v_2^i, \ldots, v_m^i\}$ tends toward minimizing the total path length $p = D(v_i, v_j)$.

For each road segment $S = (V, P)$, the connectivity is calculated as

$$\min P(S) = \sum_{(v_i,v_j)\in S} p(v_i, v_j) \quad (4)$$

where p represents the Euclidean distance between two nontopological vertices and P denotes the total road segment path length. This optimization function aligns with the traveling salesman problem (TSP), treating each nontopological vertex as a city, with the connectivity of the nontopological vertices equating to the path between each pair of cities. Notably, the number of nontopological vertices in a road segment is limited, making optimization in this TSP scenario efficient and straightforward. In conclusion, the connectivity is incorporated into the road segment, thereby enhancing the connectivity between nontopological vertices, as shown in Fig. 7.

D. Large-Scale Image Inference

When inferring large-scale remote sensing imagery larger than 512 × 512, the widely used sliding window inference technique is adopted in ISDetector inference. In the process of merging road segments within the ICS, the first step is merging intersections that are too close (Euclidean distance less than 10). Instances of road segments with an intersection over union (IoU) greater than 0.7 are merged, favoring those segments with higher confidence scores. When intersecting two patches along the same road, the proposed approach involves initially identifying all the intersections present on the road segments within the two patches. Subsequently, reconstructing and connecting the road vector graphs based on all the nontopological vertices within each segment can proceed, as shown in Fig. 8.

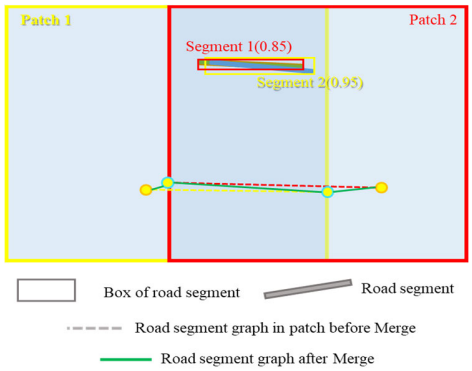


Fig. 8. Illustration of the merging strategy for the road vector graph within adjacent patches during sliding patch inference.

IV. Experiments

In order to verify the performance of the proposed IS-RoadDet framework, it was tested on the SpaceNet3 dataset and Sat2Graph dataset [12] and compared with the previous post-segmentation-based methods of DinkNet [21], Orientation [9], CoANet [11], and GAMSNet [10]; and the previous vertex-based methods of RoadTracer [14], RNGDet [13], Sat2Graph [12], TD-Road [17], Mastering [2], TopoRoad [30], GraphMapper [18], and SAM-Road [20]. To verify the validity of IS-RoadDet, the SpaceNet dataset and the Sat2Graph dataset [12] were used to conduct experiments.

A. Implementation Details

1) Framework: For the ISDetector, the common data augmentation strategy, flipping, was applied with a 0.5 probability to the training images. The images were resized to 800 × 800. The AdamW optimizer was used with a batch size of 2 and a weight decay of $2e^{-4}$. The ISDetector was trained for 12 epochs, and the learning rate was decayed by a factor of 0.1 at the ninth and the 11th epoch. The ISDetector was built on the MMDetection [43] code base. For the ICS, the threshold for the road segment instances was set to 0.25, and the threshold for the intersection heat map was set to 0.5. The buffer distance for the primitive intersections was set to 10.

2) Metrics:

a) Pixel-level metrics: The relaxed versions of precision, recall, and $F1$-score were built with different relax values. First, the road vector graph was transformed into a pixel-level road mask, controlling the road width by adding a buffer. Subsequently, the pixel-level evaluation metrics were employed to compare the positional accuracy and geometric structures of

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TABLE I
QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE SPACENET3 DATASET, WITH THE PIXEL-WISE AND TOPOLOGICAL METRICS

Modeling	Network	Relax=3			Relax=7			Relax=10			APLS
		R-P	R-R	R-F	R-P	R-R	R-F	R-P	R-R	R-F	
$G=P^S$	**DinkNet*** *2018 CVPR*	-	-	-	88.42	60.06	68.80	-	-	-	56.93
	Orientation *2019 CVPR*	73.43	56.64	63.95	90.49	71.34	79.78	93.23	74.49	82.81	64.67
	CoANet *2021 TIP*	62.43	54.37	58.12	84.08	73.21	78.27	89.00	77.82	83.04	64.57
	GAMSNet *2021 ISPRS*	72.86	70.19	71.50	84.44	81.42	82.90	90.78	87.98	89.36	67.50
$G=(V,E)$	**RoadTracer*** *2018 CVPR*	-	-	-	78.61	62.45	69.60	-	-	-	56.03
	RNGDet *2020 TGRS*	62.34	60.00	61.15	81.27	80.84	81.05	84.71	84.61	84.66	65.77
	Sat2Graph* *2020 ECCV*	-	-	-	85.93	76.55	80.97	-	-	-	64.43
	TD-Road* *2020 ECCV*	-	-	-	84.81	77.80	81.15	-	-	-	65.15
	GraphMapper* *2023 ICCV*	-	-	-	-	-	-	90.70	84.60	87.1	68.80
	Mastering* *2023 ICCV*	-	-	-	78.45	78.21	78.33	-	-	-	65.87
	TopoRoad* *2024 TGRS*	-	-	-	-	-	-	-	-	-	66.86
$G=(I,S)$	***IS-RoadDet***	**60.65**	**62.00**	**61.32**	**79.71**	**85.26**	**82.39**	**83.58**	**90.06**	**86.70**	**70.12**

* The author only provided the results, and we did not find any code that could reproduce the results, so we only report some of the quantitative results.

the road vector graph, which is defined as (5) and (7). $\hat{G}$ is the predicted road vector graph and G^* is the road graph label. For every pixel q in G^*, if there is a pixel p in $\hat{G}$ and the Euclidean distance between them is smaller than ε, then this pixel is treated as correctly retrieved. Similarly, if a pixel q in $\hat{G}$ can find a pixel p in G^* within the ε distance, then this pixel is considered to be correctly detected

$$R-P=\frac{|\left\{p|\|p,q\|<\varepsilon,\exists q\in G^*,\forall p\in\widehat{G}\right\}|}{\left|\widehat{G}\right|} \tag{5}$$

$$R-R=\frac{|\left\{p|\|p,q\|<\varepsilon,\exists q\in\widehat{G},\forall p\in G^*\right\}|}{|G*|} \tag{6}$$

$$R-F=\frac{2P\cdot R}{P+R}. \tag{7}$$

b) Graph level metric: The average path length similarity (APLS) was used to measure the similarity and topological connectivity between the ground truth and the proposal road vector graph [44]. The distance and topological similarity between the two graphs were assessed by examining the road vector graph connectivity distance among all the pairs of matched points with close distances in both the proposal graph and label graph.

B. Comparison Experiments on the SpaceNet3 Dataset

1) SpaceNet3: Following Sat2Graph [12], the ground truth of the test data in the SpaceNet3 dataset is not public [44]. The 2549 tiles (nonempty) of the original dataset were split into train (80%), test (15%), and validation (5%) datasets. Each tile was 400 × 400 pixels, which involved resizing the spatial resolution of the original tiles of 1300 × 1300 pixels to 1 m/pixel. The dataset covers four areas: Las Vegas, Paris, Shanghai, and Khartoum.

Table I provides a quantitative comparison with the results of the other state-of-the-art methods on the SpaceNet3 dataset. The IS-RoadDet framework achieves the best performance in the APLS graph connectivity metric. In terms of the pixel-wise evaluation metric, the post-segmentation methods perform better, but the large number of disconnected roads leads to poor connectivity in the APLS metric. Among the post-segmentation methods, GAMSNet obtains the best performance in the APLS metric and performs better than the vertex-based methods. This is due to the fact that when the vertex-based results are evaluated at the pixel level, a slight offset of the road vector line compared to the center of the road will result in a slight difference in the extent of the road from the pixel-level results, which may be one or two pixels, and when the length of the road accumulates to a certain point, this slight offset will be very evident in the pixel-level metrics. In terms of the pixel-wise evaluation metric, the methods based on post-segmentation are superior, but the large number of disconnected roads leads to poor connectivity in the APLS metric. The serious imbalance between the high pixel-level metrics and the low topological metrics indicates that the post-segmentation methods, in order to obtain better pixel

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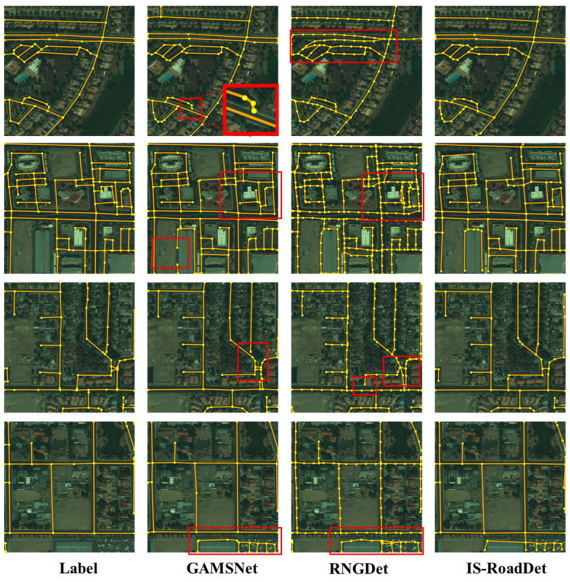


Fig. 9. Qualitative road vector graph results on the SpaceNet3 dataset. The road vector graph generation results are visualized on remote sensing images. Road vector graph (orange lines and yellow dots).

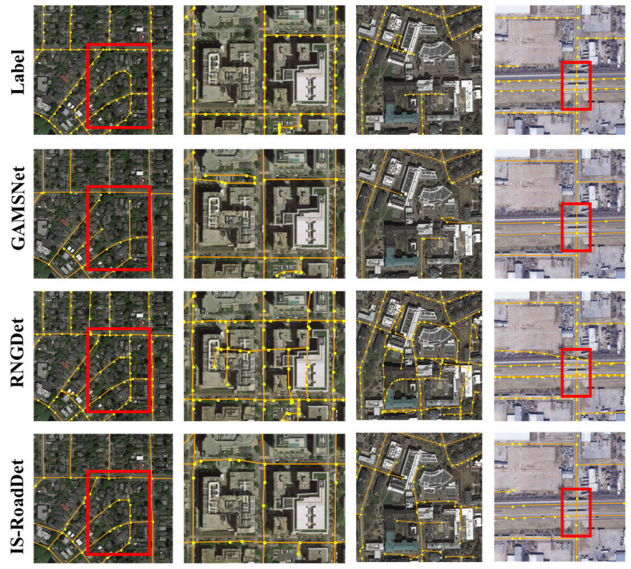


Fig. 10. Qualitative road vector graph results on the Sat2Graph dataset. The road vector graph generation results are visualized on remote sensing images. Road vector graph (orange lines and yellow dots).

segmentation results, extract many road pixels without ensuring the road topology. This supervision without road topology results in low topological metric scores. Compared to the vertex-based methods, the proposed IS-RoadDet framework obtains the best connectivity in the APLS metric and the best pixel-level performance in the R-F metric with different relax values. IS-RoadDet not only obtains a better road topology but also extracts road vectors that are closer to the real road centerline.

Fig. 9 provides the qualitative visualizations of GAMSNet, RNGDet, and IS-RoadDet on the SpaceNet3 dataset. The first column is the graph labels. The second column shows the results of GAMSNet. The third column shows the results of RNGDet. The fourth column shows the results of the proposed IS-RoadDet framework, which is able to generate a road vector graph with fewer disconnected roads and false positives. In the first row, the road vector results of GAMSNet show a disconnection at the intersection, which is due to the relatively low confidence of the road segmentation heatmap at the intersection. Meanwhile, the road vector results show incorrect topology due to the large number of nontopological vertices that do not consider the topology of the whole road segment. The road vector result of IS-RoadDet does not have disconnections or incorrect topology, which is an advantage of modeling roads using the minimum road topology unit. In the second and third rows, GAMSNet appears to show road disconnections, while the proposed method can extract the road vector completely. RNGDet shows a problem with false positive roads, and IS-RoadDet is successful in obtaining a complete and simple road vector. The comparison road vector results on the fourth row indicate that the results of IS-RoadDet are still valid for very short road segments.

C. Comparison Experiments on the Sat2Graph Dataset

1) Sat2Graph Dataset: This dataset contains 180 large-scale remote sensing imageries, each of which is of 2048×2048 pixels with 1-m spatial resolution. The 180 tiles were split into train (80%), test (15%), and validation (5%) datasets. The dataset provides remote sensing imagery focusing on 20 U.S. cities. The dataset covers downtown areas with complex backgrounds and structures, such as overpass and bridges, which is a challenging scenario. To facilitate training, the 180 imagery tiles were cropped to 1024×1024 image patches.

Table II provides a quantitative comparison with the results of the other state-of-the-art methods on the Sat2Graph dataset. IS-RoadDet achieved the best APLS metrics. Compared to the vertex-based methods, IS-RoadDet shows both the best APLS metric scores and the best pixel-level evaluation metric scores. For the APLS evaluation metric, IS-RoadDet performs 2.94% better than RNGDet, which is the best method based on $G = (V, E)$, and 9.27% better than GAMSNet, which is the best method based on $G = (P^S)$. In terms of the pixel-wise evaluation metric, the methods based on post-segmentation perform better, but the large number of disconnected roads leads to poor connectivity in the APLS metric. The serious imbalance between the high pixel-level indicators and the low topology indicators indicates that the post-segmentation methods, in order to obtain better pixel segmentation results, segment many road pixels without ensuring the road topology. This supervision without road topology results in low topology scores. In the Sat2Graph dataset, the proposed IS-RoadDet method exhibits good precision when the road width is 5 (relax = 10), which proves that the proposed method can solve the false positive problem. A similar situation occurs in the Sat2Graph dataset, with low APLS metric scores for the post-segmentation methods.

Fig. 10 provides the qualitative visualizations of GAMSNet, RNGDet, and the proposed IS-RoadDet framework on the Sat2Graph dataset. The first row is the graph labels. The second row shows the results of GAMSNet. The third row shows the results of RNGDet. The fourth row shows the results of the

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TABLE II
QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE SAT2GRAPH DATASET, WITH THE PIXEL-WISE AND TOPOLOGICAL METRICS

Modeling	Network	Relax=3			Relax=7			Relax=10			APLS
		R-P	R-R	R-F	R-P	R-R	R-F	R-P	R-R	R-F	
$G=P^S$	**DinkNet*** *2018 CVPR*	-	-	-	78.63	48.07	57.42	-	-	-	54.08
	GAMSNet *2021 ISPRS*	65.35	71.25	68.17	83.96	85.27	84.61	86.57	87.81	87.18	60.09
$G=(V,E)$	**RoadTracer*** *2018 CVPR*	-	-	-	78.00	57.44	66.16	-	-	-	57.29
	RNGDet *2020 TGRS*	52.65	71.79	60.74	74.11	86.53	79.84	77.80	89.51	83.25	66.40
	Sat2Graph* *2020 ECCV*	-	-	-	80.70	72.28	76.26	-	-	-	63.14
	TD-Road* *2020 ECCV*	-	-	-	81.94	71.63	76.27	-	-	-	65.74
	GraphMapper* *2023 ICCV*	-	-	-	-	-	-	89.6	82.6	85.6	68.00
	SAM-Road* *2024 CVPR*	-	-	-	90.47	67.69	77.23	-	-	-	68.37
$G=(I,S)$	***IS-RoadDet***	**48.12**	**71.49**	**57.32**	**72.34**	**88.29**	**79.52**	**79.49**	**91.73**	**85.17**	**69.36**

* The author only provided the results, and we did not find any code that could reproduce the results, so we only report some of the quantitative results.

proposed IS-RoadDet framework, where the proposed method is able to generate road vector graph with fewer disconnected roads and false positives. In the first to third columns, diverse occlusions, such as those caused by trees and buildings, result in disconnected roads in the output of GAMSNet. RNGDet exhibits false positive roads in complex backgrounds such as parking lots, residential areas, structurally intricate buildings, and lakes. The proposed IS-RoadDet framework demonstrates the capability to extract continuous roads while effectively suppressing the generation of false positive roads. In the fourth column, GAMSNet generates a road vector graph with incorrect topology, whereas the definition of the upper and lower road segment instances in the proposed approach ensures the generation of a road vector graph with correct topology.

From the pixel-level metrics and topological metrics on the two datasets, it can be seen that the proposed method can effectively solve the problem of false positive roads and disconnections, balancing the accuracy of road extraction and road topology.

D. Large-Scale Image Inference

Table III gives the quantitative results for the R128 image from the Sat2Graph dataset, where IS-RoadDet shows a certain degree of improvement in the APLS metric. IS-RoadDet is 10.99% higher than the best pixel-level modeling approach and 5.37% higher than the best vertex-based approach in the APLS metric. Fig. 11 illustrates the road vector graph detection results on the R128 image of the Sat2Graph dataset by large-scale image inference with IS-RoadDet. In order to visually compare the results of the different methods, we zoom in on three local areas to compare the details in Fig. 12. From the overall large-scale image, the city roads are basically extracted completely by IS-RoadDet. In the top-right corner of the large-scale image, the distribution of buildings

TABLE III
QUANTITATIVE RESULTS OF THE DIFFERENT METHODS ON THE R128 IMAGE OF THE SAT2GRAPH DATASET

Modeling	Network	R-P	R-R	R-F	APLS
$G=P^S$	**DinkNet** *2018 CVPR*	90.01	87.27	88.62	67.15
	GAMSNet *2021 ISPRS*	89.93	89.04	89.49	68.49
$G=(V,E)$	**RNGDet** *2020 TGRS*	80.72	90.24	85.22	74.13
	Sat2Graph *2020 ECCV*	85.52	88.82	87.14	70.91
$G=(I,S)$	**IS-RoadDet**	**78.32**	**90.40**	**83.93**	**79.50**

is very regular, and the road network is basically error-free. In the bottom-left corner, the distribution of buildings and roads is more scattered, resulting in some disconnections in the road network. In Area 1, on a triangular intersection, both GAMSNet and RNGDet do not detect a short road, while IS-RoadDet detects the three road segments completely. In Areas 2 and 3, GAMSNet has a more serious missed detection problem, while RNGDet shows false positives due to the complex background, and IS-RoadDet performs the best of all.

Table IV lists the quantitative results for the R88 image from the Sat2Graph dataset, where it can be seen that IS-RoadDet shows a certain degree of improvement in the APLS metric. IS-RoadDet is 9.53% higher than the best pixel-level modeling approach and 10.48% higher than the best vertex-based approach in the APLS metric Fig. 13 shows the road vector graph detection results on the R88 image of the Sat2Graph dataset by large-scale inference with IS-RoadDet. In the same way, Fig. 14 shows three detail images for a comparison with GAMSNet and RNGDet, which are the best post-segmentation

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Fig. 11. Large-scale mapping results of IS-RoadDet on the R128 image of the Sat2Graph dataset.

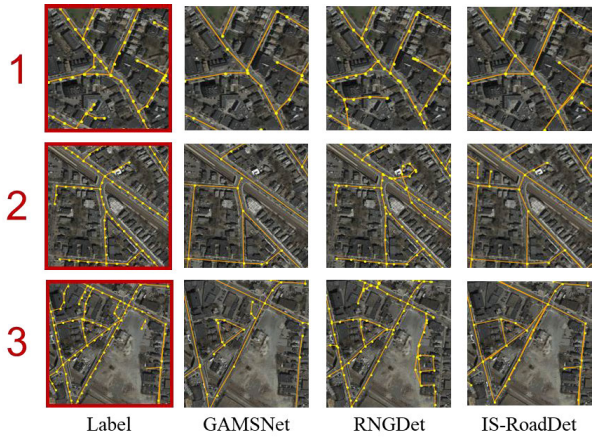


Fig. 12. Visual results for the three areas in the R128 image of the Sat2Graph dataset.

method and vertex-based method from the quantitative results, respectively. Fig. 13 shows a classical urban image, which consists of dense built-up areas and overpasses. The road vector extraction results for the built-up area and the overpass area in the upper half of the urban area are complete, but one overpass at the bottom of the image has an underdetection problem due to lane line inconsistencies resulting in juxtaposed lane lines. In Area 1 of Fig. 14, a thin river is detected as a road by RNGDet, generating false positives, whereas IS-RoadDet has no such problem. In Areas 2 and 3, GAMSNet and RNGDet generate roads with incorrect topological connectivity in the case of the overpasses and overpass-like roads, while IS-RoadDet obtains roads with correct topological connectivity.

From the results of inference on large-scale images, the road extraction outcomes in urban areas exhibit minimal visual errors. Even when substantial occlusions occur due to buildings and roadside trees along certain road segments, the extracted road vector graph maintains strong connectivity and coherence.

TABLE IV

QUANTITATIVE RESULTS OF THE DIFFERENT METHODS ON THE R88 IMAGE OF THE SAT2GRAPH DATASET

Modeling	Network	R-P	R-R	R-F	APLS
$G=P^S$	**DinkNet** *2018 CVPR*	85.35	89.47	87.36	72.10
	GAMSNet *2021 ISPRS*	84.88	90.57	87.63	73.13
G= (V, E)	**RNGDet** *2020 TGRS*	75.68	91.91	83.01	72.18
	Sat2Graph *2020 ECCV*	76.15	89.37	82.23	65.93
G= (I, S)	**IS-RoadDet**	**79.31**	**89.30**	**84.01**	**82.66**

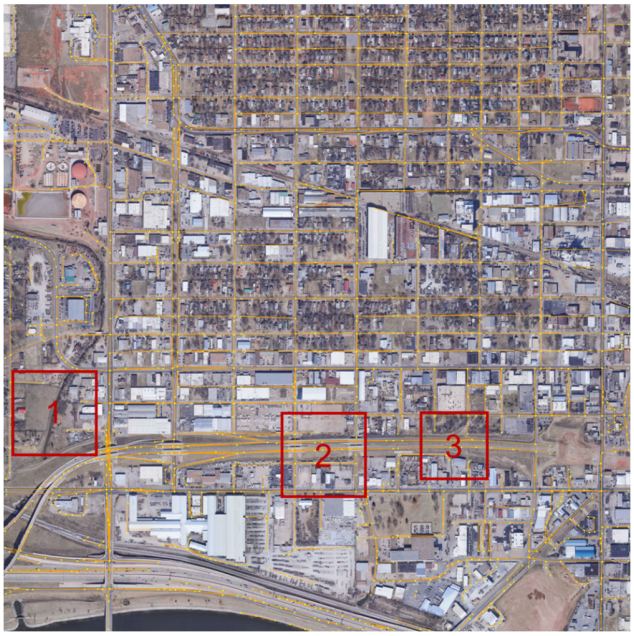


Fig. 13. Large-scale mapping results of IS-RoadDet on the R88 image of the Sat2Graph dataset.

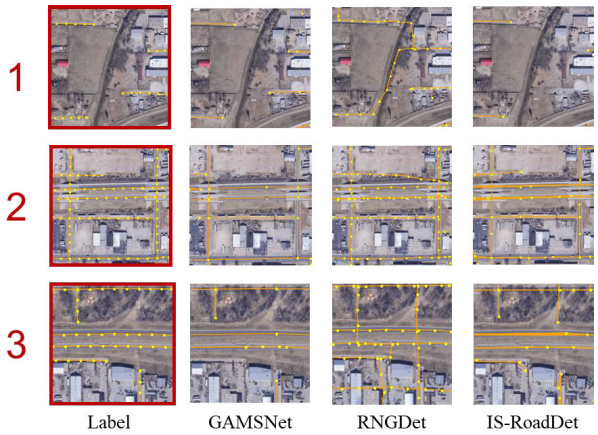


Fig. 14. Visual results for three areas in the R88 image of the Sat2Graph dataset.

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TABLE V
ABLATION STUDY OF FOR THE ICS USING IS-ROADDET ON THE SPACENET3 AND SAT2GRAPH DATASETS

	ICS	APLS
SpaceNet3	✗	62.91
	✔	70.12
City-Scale	✗	59.60
	✔	69.36

E. Discussion

1) Effectiveness of the ICS: The ICS is an important part of IS-RoadDet that is mainly used to connect road segments and intersections to generate the road vector graph. In order to investigate the effectiveness of the designed ICS, we compared it with the simple strategy of connecting all the road segments. Table V illustrates the effectiveness of the ICS on both datasets. To assess the isolated performance of the ICS, we exclusively utilized the nontopological vertex connectivity derived from the road instance segmentation results to generate the road vector graph, omitting the part where the ICS optimizes the topological vertex connectivity. The results from the ablation experiments clearly indicate the significant role played by the ICS in enhancing the connectivity performance for nontopological vertices. This underscores that the unique minimum distance connectivity attribute of road segments, serving as the smallest topology unit of the road vector graph, can be leveraged for nontopological vertex connectivity while effectively addressing occlusion issues. Consequently, the nontopological vertices embedded in the road segments have been shown to be highly suitable for detection. Integrating this with the distinct characteristics of the topological vertices discussed in Section I, where we distinguish between topological and nontopological vertices in the proposed modeling approach $[G = (I, S)]$ and embed nontopological vertices into the road segments, further reinforces the suitability of the proposed road vector graph modeling $[G = (I, S)]$ approach. This validation emphasizes the superiority of the proposed approach in road vector graph generation. Because the minimum topological attribute characteristics of road segments can be effectively utilized to optimize the heuristic ICS, this also demonstrates the validity and effectiveness of the proposed instance definition approach for road segments.

2) Uniqueness of the Topological Vertices (I): In road vector graph modeling, in comparison to nontopological vertices, topological vertices not only govern the road vector graph's topology but also possess a more explicit spatial location attribute. To validate this assertion, four types of vertices (intersection, endpoint, corner point, and trivial point) were employed in intersection detection to supervise the detection of all the vertices, anticipating that the topological vertices with clear definitions and attributes should be readily detectable. A detected vertex was considered correct if it was within a 5-m Euclidean distance of the corresponding vertex in the road vector graph labels. The uniqueness of the topological vertices was assessed through the performance of the detected topological and nontopological vertices on the SpaceNet3 dataset. The recall and precision for the topological vertices are 53.72% and 62.06%, respectively, and the recall and precision for the nontopological vertices are 14.55% and 15.63%, respectively. As anticipated, the nontopological vertices are challenging to detect, given their lack of precise spatial location attributes. Consequently, the nontopological vertices are unsuitable for representation as points in detection. Thus, distinguishing between the topological and nontopological vertices in modeling and embedding nontopological vertices in road segments can be considered a reasonable approach.

3) Intermediate Results of the ISDetector: Due to the two-stage nature of the proposed method, the instance detection results and segmentation outcomes of the ISDetector can be directly examined. In addition, as the extraction results for road segments and the segmentation outcomes for intersections significantly influence the final results, the intermediate results were evaluated. The commonly used metrics for instance detection were employed to quantitatively showcase the performance, revealing that a satisfactory level of instance detection can be achieved. From a visual inspection, the road segments are effectively identified, and there are no issues in distinguishing between them. Following the proposed modeling approach, instance-level road segment detection is realized and segmentation of the topological vertices is also clearly defined. The intermediate results are shown in Fig. 15. In order to evaluate the performance of the ISDetector, the segmentation results for the instances of road segments were evaluated on the SpaceNet3 dataset. The bbox mAP50 metric score is 53% and the seg mAP50 metric score is 28.20%, so the quantitative results indicate that the segmentation results for the road instances still have a lot of room for improvement. However, although the metric scores of the instance segmentation are not high, the connectivity metrics of the roads are high, which is mainly due to the fact that the ICS eliminates a large number of false positive instances.

4) Region- and Query-Based Detector: The region-based instance detection method is a stable and commonly used approach for instance detection. However, when dealing with road segment instances characterized by intricate geometric structures, significant length variations, and interwoven topological relationships, region-based detection localization often struggles to effectively distinguish between road segments. Therefore, the query-based instance detection method was adopted to address the challenge of ineffective discrimination of road instances. To validate this approach, experiments using the region-based instance detection method of Mask-RCNN were conducted, and we compared the road extraction results with those of the query-based instance detection method of Mask2Former [40]. The results in Table VI indicate that the query-based instance detector performs better in detecting road instances, demonstrating the limitation of region-based instance detection methods for road segment instances.

5) Model Complexity and Time Complexity Analysis: Representative Road graph extraction methods are selected to evaluate model complexity and computational time in Table VI.

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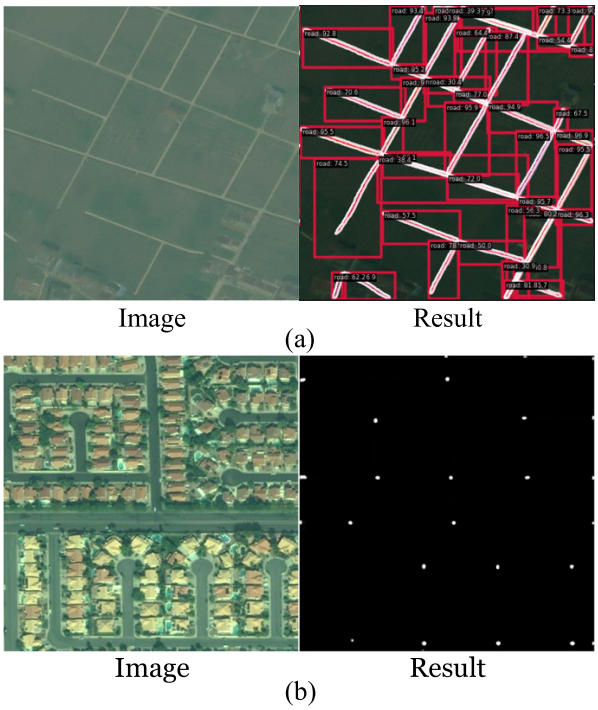


Fig. 15. Visual results of the ISDetector. (a) Instance segmentation of road segments. (b) Topological vertex segmentation of intersections.

TABLE VI

QUANTITATIVE COMPARISON OF THE REGION- AND QUERY-BASED IS-ROADDET ON THE SPACENET3 AND SAT2GRAPH DATASETS

Dataset	Different ISDetectors	R-P	R-R	R-F	APLS
SpaceNet3	IS-RoadDet (Region-based)	79.67	83.82	81.69	68.74
	IS-RoadDet (Query-based)	**79.71**	**85.26**	**82.39**	**70.21**
Sat2Graph	IS-RoadDet (Region-based)	71.81	83.80	77.34	67.89
	IS-RoadDet (Query-based)	**72.34**	**88.29**	**79.52**	**69.36**

TABLE VII

MODEL COMPLEXITY AND TIME COMPLEXITY OF THE PROPOSED METHOD AND SOME OF THE STATE-OF-THE-ART METHODS

Methods	Params.(M)	GFLOPs
DinkNet	217.79	121.21
GAMSNet	29.33	58.32
CoANet	59.04	276.70
RoadTracer	27.30	73.34
Sat2Graph	55.29	373.58
RNGDet	66.56	430.99
IS-RoadDet	71.43	382.07

Post-segmentation methods generally exhibit lower model and time complexity. In contrast, methods that directly generate road vectors tend to be more computationally intensive, particularly because both RNGDet and IS-RoadDet are built upon the transformer architecture.

V. CONCLUSION

In this article, IS-RoadDet has been proposed to generate a road vector graph with a road vector graph modeling method, denoted as $G = (I, S)$, specifically designed for VHR remote sensing imagery. Our key innovation involves partitioning the road network into discrete road segment instances, each encapsulating the connectivity of both topological and nontopological vertices to intersections and road segments, respectively. The road segment, serving as the smallest unit of road topology, inherently addresses issues such as disconnected and false positive roads. In addition, the distinctive smallest path attribute of road segments is leveraged in the ICS to enhance the connectivity of nontopological vertices. IS-RoadDet demonstrated competitive results on both the SpaceNet3 and Sat2Graph datasets when compared to the existing modeling methods. In the future, considering road segments as the smallest units for topology, attributes, and changes, we envision extending the proposed road vector graph modeling method to tasks such as road vector graph attribute detection (e.g., width, levels, lanes, and disaster roads) and even road vector graph updating, which is a challenging task with the prior modeling methods. We hope that this contribution will offer valuable insights into the field of road extraction from high-resolution remote sensing imagery.

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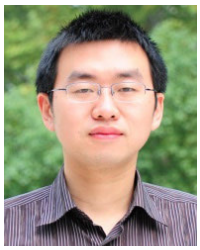
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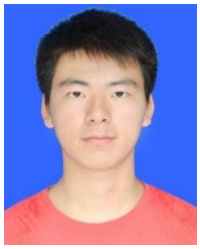
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